MACHINE LEARNING PORTFOLIO / LABS 2-10

Fingerprint spoofing classification

Project report | Ali Badran | October 2026

This project compares classifiers for distinguishing fake and genuine fingerprints using six extracted numerical features. The work starts with data exploration and ends with calibrated score-based decisions on a separate evaluation set. It is a study of the supplied feature dataset, not a complete fingerprint-image recognition system.

The methods are multivariate Gaussian (MVG), logistic regression (LR), support vector machines (SVM), and Gaussian mixture models (GMM).

Main result. Five-fold calibration validation selects the calibrated GMM by actual detection cost. On the separate evaluation set it reaches **actual DCF 0.2423** and **minimum DCF 0.2331**. Three-system fusion performs better there, with actual DCF 0.2203, but the delivered choice stays fixed from validation.

Data and experimental protocol

Partition	Samples	Purpose
Model training	4,000	Fit classifier parameters and preprocessing, where used
Validation / calibration	2,000	Choose hyperparameters; assess calibration in five held-out folds
Final evaluation	6,000	Score fixed models and calibration maps; no fitting or reselection

The supplied training file contains 6,000 samples and is split once using seed 0: a NumPy RandomState permutation, the first two thirds for model training, and the rest for validation. Fake is class 0 and genuine is class 1. Class counts are nearly balanced, but the application metric uses genuine prior 0.1.

How performance is measured

A false positive accepts a fake fingerprint; a false negative rejects a genuine one. FPR and FNR are their class-specific rates. With genuine prior 0.1 and unit mistake costs, the normalized detection cost is:

```
normalized DCF = FNR + 9 * FPR
```

Actual DCF uses the Bayes threshold on log-likelihood-ratio (LLR) scores: log(9), approximately 2.1972. **Minimum DCF** sweeps thresholds using the assessment labels. It measures available discrimination and is a diagnostic lower bound, not the achieved application cost. A cost of 1 matches the best constant decision.

Final candidate classifier settings are selected by validation minimum DCF. After calibration, the delivered method is selected by validation actual DCF. All final evaluation results are reported without changing that choice.

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What the features reveal

The six features contain different kinds of information. Features 1-2 mainly differ in spread. Features 3-4 have substantially shifted class means. Genuine features 5-6 have two visible modes each, and their joint scatter plot shows four clusters; fake samples have a more central distribution.

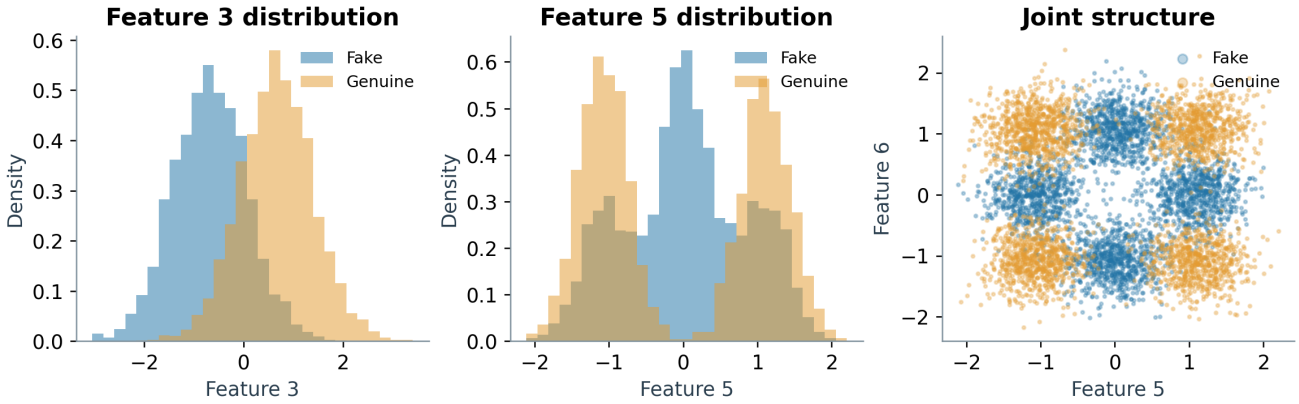


Figure 1. Descriptive plots use the 6,000-sample training file only. Density histograms use shared bins. The scatter plot shows visual groups, not fitted mixture components.

These shapes help explain the later results: a mean-based linear boundary misses some variance and multimodal information. A single Gaussian cannot reproduce the two genuine peaks in features 5-6. Weak within-class linear correlations also do not establish independence.

PCA and LDA

Principal component analysis (PCA) keeps directions with the greatest variance. Linear discriminant analysis (LDA) uses labels to find directions with strong class-mean separation relative to within-class spread.

Retained dimensions	Pooled variance retained	PCA + LDA validation error
1	23.99%	187 / 2,000 (9.35%)
2	40.94%	185 / 2,000 (9.25%)
4	74.29%	185 / 2,000 (9.25%)
6	100.00%	186 / 2,000 (9.30%)

The variance column describes exploratory PCA fitted on the whole training file. Classification PCA and LDA are re-estimated using only the 4,000 model-training samples. LDA without PCA makes 186 validation errors, so reducing to two, three, or four dimensions improves the count by just one.

Keeping all six principal components only rotates the data and preserves distances. Dropping components can discard useful class information. The results provide little reason to reduce dimensionality for the final models, so the selected LR, SVM, and GMM classifiers keep all six original features.

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From linear rules to curved boundaries

Full-covariance MVG fits one Gaussian per class. Tied MVG shares a covariance and gives a linear rule. Naive Bayes uses a separate diagonal covariance for each class. At equal priors, all-feature MVG makes 7.00% validation errors, Naive Bayes 7.20%, and tied MVG/LDA 9.30%. Those error rates use a different application from the cost comparison below.

Validation configuration	minDCF	actDCF
MVG, all six original features	0.2629	0.3051
Naive Bayes, all six original features	0.2570	0.3022
Tied MVG, PCA with four dimensions	0.3610	0.4031
Linear LR, lambda=0.01	0.3611	0.4568
Quadratic LR, lambda=0.03162	0.2436	0.4972
Degree-2 SVM, C=0.03162	0.2455	0.4674
Degree-4 SVM, C=0.001	0.1775	0.3988
RBF SVM, gamma=exp(-2), C=31.62	0.1755	0.4216

Costs use prior 0.1 and unit costs. Each row shows a configuration selected by minimum cost within the displayed family; its actual cost is evaluated at that same setting.

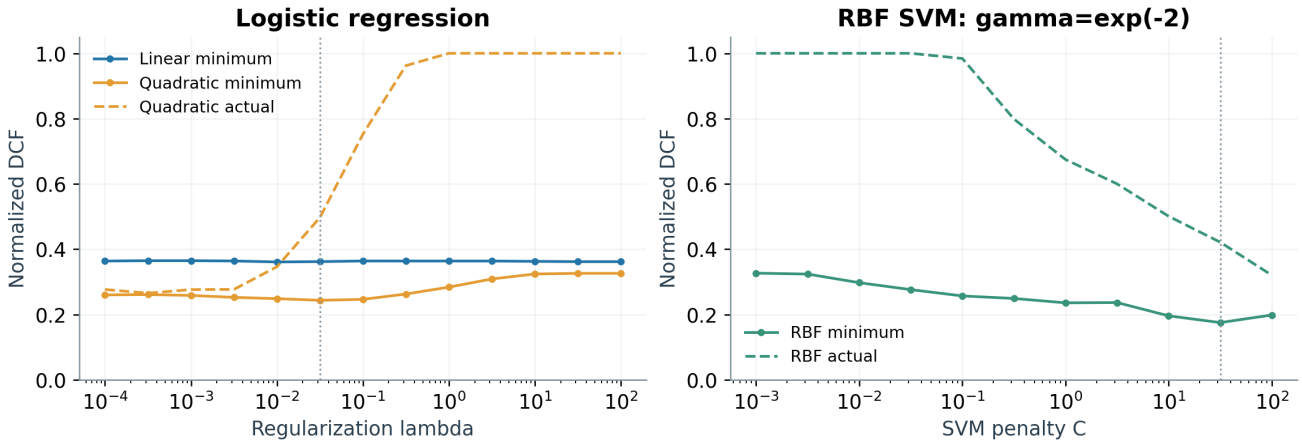


Figure 2. Validation parameter grids. Vertical dotted lines mark the selected quadratic LR and RBF SVM settings. Minimum cost and actual cost can favor different regularization values.

Quadratic LR expands six features to 42 terms, including squares and pairwise products. Polynomial and radial basis function (RBF) kernels can use curved or localized boundaries. Their lower minimum costs fit the observed class structure. RBF and degree-4 SVM costs differ by only about 0.002, so this split does not establish a reliable advantage between them.

Good separation is not sufficient: the selected quadratic LR and RBF SVM have large actual/minimum gaps. Their scores need calibration before the Bayes threshold can use their discrimination effectively.

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Mixtures fit the class structure better

A Gaussian mixture models each class as a weighted sum of Gaussian components. Expectation-maximization (EM) alternates soft component assignments and weighted parameter updates. Linde-Buzo-Gray (LBG) initialization grows the mixture by splitting components along their largest-variance direction, then running EM again.

The project uses full covariances, an eigenvalue floor of 0.01, split displacement 0.1, and an average log-likelihood improvement stopping threshold of 1e-6. Fake and genuine models independently test 1, 2, 4, 8, and 16 components: ten class models produce 25 classifier combinations.

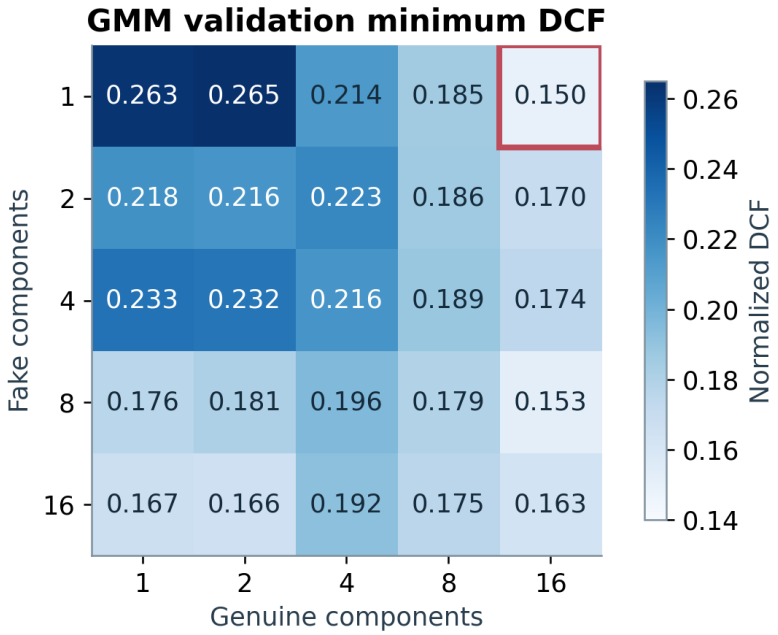


Figure 3. Minimum DCF on the 2,000 validation samples at prior 0.1. The outlined cell selects one fake component and 16 genuine components.

The winning combination reaches minimum DCF 0.1495 and actual DCF 0.2055. Restricting both classes to the same count would miss it: 16/16 gives minimum cost 0.1631. A single component per class reproduces full MVG, with minimum cost 0.2629. More components are not guaranteed to improve validation performance.

Fixed candidate for calibration	minDCF	actDCF
GMM: 1 fake / 16 genuine components	0.1495	0.2055
Quadratic LR: lambda=0.0316228	0.2436	0.4972
RBF SVM: gamma=exp(-2), C=31.6228	0.1755	0.4216

These three previously selected models are kept fixed for calibration and evaluation. Each still uses parameters fitted on the original 4,000 model-training samples; calibration does not retrain the base classifiers.

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Calibration and model selection

Calibration is unregularized, prior-weighted logistic regression on scores, trained with prior 0.1. For one system it learns a scale and offset. Fusion applies the same method to all three score rows jointly. The training-prior log odds are subtracted once to produce an LLR-like score.

```
calibrated score = w @ input_scores + corrected_offset
```

Five-fold calibration uses interleaved folds i::5. Every held-out score is produced by a map trained on the other four folds; scores stay aligned with their original labels. Actual cost on these pooled scores chooses the delivered method.

Candidate	Raw minDCF	Raw actDCF	Five-fold minDCF	Five-fold actDCF
GMM	0.1495	0.2055	0.1575	0.1677
Quadratic LR	0.2436	0.4972	0.2486	0.2721
RBF SVM	0.1755	0.4216	0.1685	0.1894
Fusion	-	-	0.1531	0.1708

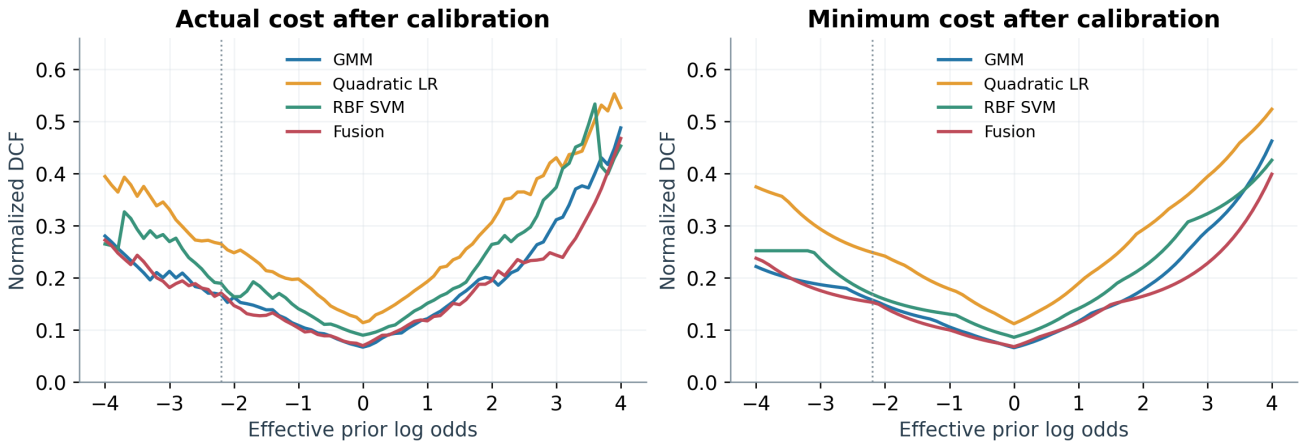


Figure 4. Five-fold calibration validation. The vertical dotted line marks genuine prior 0.1. All four methods are scored on the same 2,000 validation samples.

Delivered choice: calibrated GMM. Its actual validation cost is 0.1677, slightly below fusion at 0.1708. Fusion has the lowest minimum cost, 0.1531, but the selection rule uses achieved actual cost after calibration.

All three individual systems improve at the target prior. Separate fold maps can change global sample ordering, so pooled minimum cost can differ from the original minimum cost. A single positive-slope affine map preserves ordering on a new evaluation set.

After selection, each final map is fitted on all 2,000 validation samples. The delivered choice and every map are fixed before opening the evaluation file. These folds hold out calibration labels; the base hyperparameters were selected using the same development labels earlier, so this is not fully nested model validation.

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What happened on final evaluation

The separate evaluation file has 6,000 samples: 3,010 fake and 2,990 genuine. It is scored using the original fitted base models and the final calibration maps, with no model fitting or selection on these samples.

Candidate	Raw minDCF	Raw actDCF	Final minDCF	Final actDCF
GMM	0.2331	0.2465	0.2331	0.2423
Quadratic LR	0.3515	0.4942	0.3515	0.3896
RBF SVM	0.2622	0.4026	0.2622	0.2889
Fusion	-	-	0.2139	0.2203

The delivered GMM achieves actual DCF 0.2423 against minimum DCF 0.2331: a gap of 0.0092, or 3.94% of minimum cost. Its confusion matrix gives 42 accepted fakes and 349 rejected genuine samples (FPR 1.40%, FNR 11.67%). Rows below are predictions; columns are true labels.

	Actual fake	Actual genuine
Predicted fake	2,968	349
Predicted genuine	42	2,641

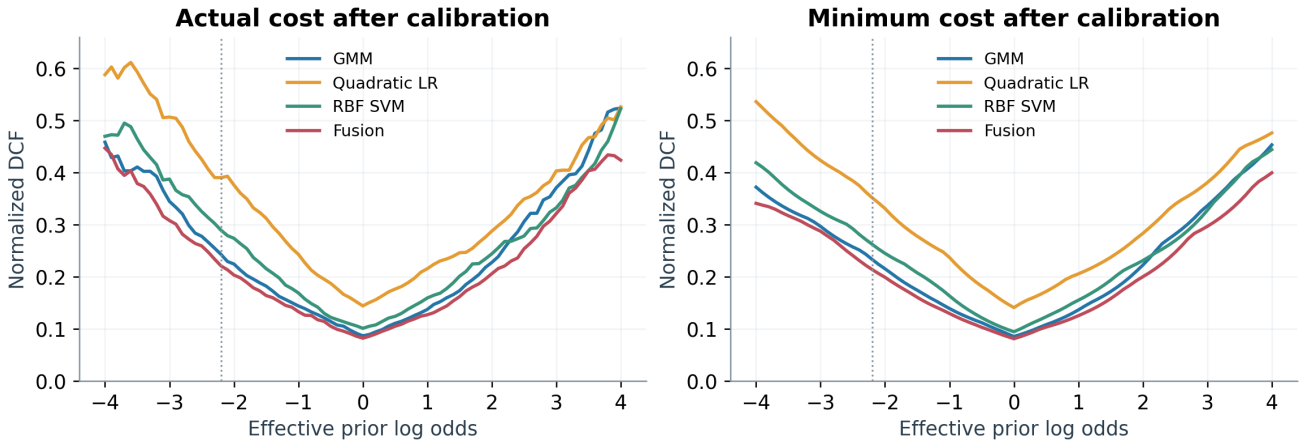


Figure 5. Fixed calibrated models on final evaluation. Across the plotted 81 prior settings, fusion has the lowest minimum cost everywhere and the lowest actual cost at 79 points; GMM wins actual cost at two.

Fusion would have been better on these data: its actual cost is 0.2203. GMM remains the strongest individual system, and stays the delivered method because validation selected it. Changing that choice now would use evaluation labels for model selection.

Calibration substantially helps LR and SVM at the target prior. Its GMM benefit is smaller, and some extreme-prior decisions worsen. At prior 0.5, GMM and fusion actual costs are 0.0870 and 0.0823; at prior 0.9, they are 0.2557 and 0.2203. All calibrated candidates stay below the constant-decision baseline on this grid.

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Model audit and limitations

The optional audit checks 52 logistic-regression configurations: 13 lambda values each for full linear, reduced linear, weighted linear, and quadratic models. The 39 full-training models are restored. The 13 reduced models are reproduced on their original 80 training samples before evaluation access. No model is fitted using evaluation data.

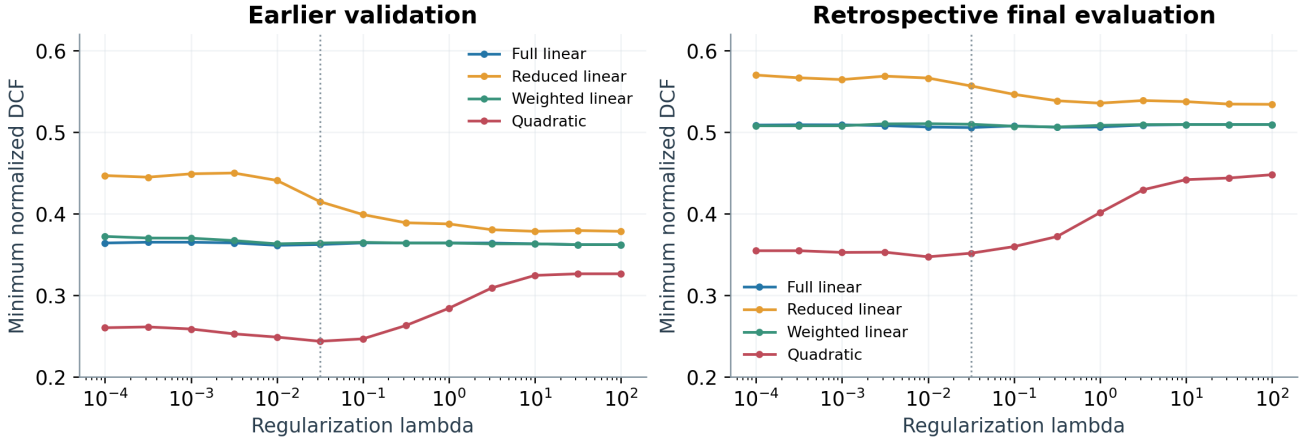


Figure 6. Minimum-cost audit at prior 0.1. The dotted line marks the selected quadratic lambda. Evaluation curves are retrospective diagnostics and do not change the delivered system.

Quadratic LR setting	Evaluation minDCF	Interpretation
Selected: lambda=0.0316228	0.3515	Second among the 52 settings
Retrospective best: lambda=0.01	0.3471	Lowest cost in the evaluated grid

The selected cost is 1.27% above the retrospective optimum. This audit compares minimum DCF only; it does not recalibrate all 52 models.

Interpretation. Nonlinear models and mixtures use information missed by mean-based linear rules. Calibration makes LR and SVM scores more useful for the target decision. GMM performs worse than its validation estimate. Fusion gives lower evaluation cost. Quadratic LR regularization was close to the best evaluated grid setting.

Limits and next steps

Results use one dataset and one model-training split. Hyperparameter selection reuses development labels; the small GMM/fusion validation difference does not establish a stable ranking. Repeated or nested development splits could assess selection stability. A revised delivered choice needs new independent evaluation data.

Reproduce the final analysis. Install requirements.txt, keep the lab folders together, and restore the local inputs and model caches supplied in the portfolio download. From the repository root:

```
python lab10/lab10.py --project --save --no-show
```

Evidence: project implementations in lab02-lab10; saved model/score archives in lab07-lab10/results/project_models.npz; configuration grids and run commands in README.md. Figures in this report are drawn from those local data and results. Project tasks and inputs originate from the supplied Machine Learning laboratory material; professor source programs and assignment PDFs are not reproduced.